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Introduction to Red Teaming

Chapter Objectives

After reading this chapter and completing the exercises, you will be able to do the following:

- Understand what red teaming is and its importance
- Identify common red team frameworks and methodologies
- Explain types of red team engagements

What Is Red Teaming?

In the preceding chapter, we dug into the history of red teaming and its roots in 1960s military war games. Then we followed along as it evolved into modern-day red teaming. Red teaming can take on a variety of forms because it can be applied in a variety of roles and even outside of cybersecurity.

Let's take a moment to outline what red teaming is. It is a broader, multi-layered approach that encompasses a wide variety of tactics, techniques, and procedures (TTPs). It utilizes not only TTPs from real-world adversaries to test technical security controls but also the human element, processes, and the overall security posture of an organization. Red teaming offers an adversarial perspective essential for testing security assumptions and evaluating the effectiveness of the people, processes, and technologies employed to defend an environment.

Any offensive security testing, such as a penetration test, is often misinterpreted as red teaming. They both play important roles in cybersecurity, but they tackle different goals and use different approaches.

The goal of a penetration test is to find as many vulnerabilities as possible in a controlled manner, typically within a defined scope and timeframe. The objective is solely to exploit the system(s) and document weaknesses for remediation. It does not challenge the detection and response capabilities, nor does it evaluate the involvement of people or processes in preventing such attacks.

Red teaming generally focuses on achieving a particular objective, such as gaining access to sensitive data or compromising critical infrastructure. It is designed to challenge the detection and response capabilities of an organization to assess how well they can identify, respond, and contain an ongoing attack over an extended period from persistent threats. The emphasis here is on creatively executing multi-step attacks to achieve the objective while avoiding detection.

Red teaming comes in many different forms, and we will dive into the wide variety of engagements later in this chapter. However, let's discuss

the importance of red teaming first.

The Importance of Red Teaming

Red teaming is an essential component of a robust cybersecurity strategy, offering a proactive approach to identifying and addressing weaknesses in an organization's defenses. By emulating real-world adversarial tactics, techniques, and procedures, red teaming provides unparalleled insights into an organization's resilience against cyber threats. Its importance can be broken down into several key aspects: identifying critical vulnerabilities, challenging assumptions, and emulating industry-specific relevant threats.

Identifying Critical Vulnerabilities

Red team engagements are uniquely designed to uncover critical vulnerabilities by adopting a holistic and multi-dimensional approach to security testing. Unlike narrowly focused penetration tests that typically target specific systems or applications, red teaming evaluates an organization's entire security ecosystem. This comprehensive methodology ensures that no critical weak points are overlooked, providing actionable insights into the organization's overall risk posture.

Holistic red teaming ensures that security measures are not only robust in isolation but also resilient when tested against complex, multi-vector attacks. This level of insight is invaluable for organizations striving to defend against sophisticated adversaries and continuously improve their security defenses.

Challenging Assumptions

Another significant benefit of red teaming is its ability to challenge existing assumptions about an organization's security posture by adopting an adversarial perspective. Security measures are often designed and implemented based on theoretical scenarios, compliance requirements, or past experiences. While these approaches provide a foundation, they may overlook evolving threats, misjudge the ingenuity of real attackers, or rely too heavily on untested assumptions.

By simulating real-world adversarial behavior, red teams put these assumptions to the test, identifying gaps or weaknesses that would otherwise remain hidden. For example, an organization may assume that multifactor authentication (MFA) is sufficient to secure sensitive accounts. A red team, however, might exploit social engineering tactics to bypass MFA, such as phishing for authentication tokens or intercepting SMS-based codes. These exercises highlight potential flaws in processes that, while theoretically robust, may fail under real-world conditions.

This validation is critical for ensuring that security controls perform as expected, not only in controlled environments but also under the stress of actual attacks. It shifts the focus from the dreaded "checkbox security" of meeting regulatory or compliance requirements to actually building security systems that are truly resilient against modern threats.

Ultimately, this adversarial perspective promotes a more adaptive and realistic approach to security. It encourages organizations to constantly

question their defenses, refine their assumptions, and stay ahead of evolving threats, making their security posture more robust and resilient.

Emulating Industry-Specific Threats

The specific threats an organization will likely face can vary depending on the industry. For example, a financial organization is often targeted by highly specialized groups, such as those involved in banking fraud or targeting payment systems. In contrast, healthcare organizations often face threats from ransomware groups seeking to disrupt critical systems or steal patient data. A tailored red team engagement might involve simulating ransomware propagation or exfiltration of electronic health records, testing not just the technical defenses but also the response protocols and business continuity plans. The exact TTPs are replicated in the red team engagements to test the relevant defenses that the organization is most likely to encounter. By replicating the exact threats an organization is likely to encounter, red teaming ensures that security measures are aligned with the realities of the threat landscape.

Red teaming provides an opportunity for organizations to gain insight into how effectively they can detect, respond to, and recover from real-life cyber attacks. The emulation of adversarial attacks gives the security operations center (SOC) the ability to evaluate current detections and alerts. Creating new detections and alerts while having the opportunity to retest and tune them is immeasurable.

Red teaming is a powerful tool for organizations to strengthen their security posture proactively. It provides a real-world lens through which to evaluate vulnerabilities, challenge assumptions, and test defenses comprehensively. By tailoring engagements to industry-specific threats and focusing on detection, response, and recovery, red teaming enables organizations to anticipate and mitigate risks effectively, fostering resilience in an ever-evolving threat landscape.

Red Team Frameworks and Methodologies

In the ever-evolving landscape of cybersecurity, red teams play a crucial role in proactively identifying and mitigating vulnerabilities before malicious actors exploit them. To effectively emulate real-world adversaries and provide meaningful assessments, red teams rely on established frameworks that guide their operations and enhance communication with clients. These frameworks offer a standardized approach to understanding adversary TTPs, enabling red teams to conduct more realistic and targeted attacks. Moreover, they bridge the gap between offensive and defensive security perspectives by providing a common language for discussing threats and vulnerabilities. This shared understanding facilitates collaboration between red teams, blue teams, and leadership within an organization. This collaboration ultimately results in more effective detection engineering and overall improvement in an organization's security posture.

The sections that follow describe the frameworks that represent some of the most common and valuable resources used by red teams today.

MITRE ATT&CK Framework

The MITRE ATT&CK Framework is a comprehensive knowledge base that systematically catalogs adversary TTPs observed across the kill chain. It is recognized across the industry and is a standard for effectively communicating about cyber attacks. It was developed by the MITRE corporation and is regularly updated to reflect the evolving threat landscape. It is utilized across organizations, such as the red team, threat intel, and SOC teams, to ensure they understand the risk and address it.

The ATT&CK Framework is organized into a number of matrices, each one focusing on a different domain. They are tailored for each of the domains: Enterprise, ICS, and Mobile. Red teams leverage the framework to emulate real-world adversary behavior during engagements. It also aids in mapping red team activities to specific TTPs, which facilitates the detailed analysis of detection engineering and the measurement of security control effectiveness.

Unified Kill Chain

The Unified Kill Chain (UKC) is a conceptual framework that is the evolution of a number of previous efforts. It addresses the limitations of previous models through the integration of their strengths in a unified approach. It is designed to provide a holistic view of adversary behavior across the entire attack lifecycle.

Unlike the MITRE ATT&CK Framework, which we previously mentioned, the Unified Kill Chain focuses on both the technical and strategic aspects of cyber attacks. It incorporates techniques from initial compromise, persistence, lateral movement, and impact while aligning with broader risk management and operational objectives. It focuses on end-to-end coverage of the attack. It ties specific technical actions to strategic goals for the attacker. This allows an organization to understand how these TTPs contribute to the broader adversary goals in isolation. The flexibility of the kill chain is that it complements other frameworks like MITRE ATT&CK, providing a unified narrative of adversarial campaigns. Lastly, it is utilized by red teams to emulate sophisticated attack sequences, helping organizations test their security controls in a holistic manner.

TIBER-EU

Threat Intelligence-Based Ethical Red Teaming (TIBER-EU) is a framework developed by the European Central Bank (ECB) for financial institutions in the European Union that provide core financial infrastructure. The main emphasis is on the use of threat intelligence to tailor exercises to specific threats faced by an organization. The other focus of the framework is to provide a consistent and comprehensive approach to red teaming, ensuring that exercises are conducted to a high standard. This approach enables the results to be comparable across institutions. The result of the exercise is focused on increasing the cyber maturity and resilience of the organization; this is done through identifying the strengths and weaknesses of the financial institution. The participants include five teams in an exercise:

- **Threat Intelligence:** A third-party team that performs reconnaissance and analysis of the financial institution. This team provides a comprehensive list of possible threat actors and their TTPs.
- **Blue Team:** The financial institution's internal team, which is responsible for the prevention, detection, and response capabilities.
- **Red Team:** A third-party team that carries out the emulated attack against the financial institution.
- **White Team:** Trusted agents internal to the financial institution.
- **TIBER Team:** The internal team that is responsible for ensuring the test meets the requirements of the TIBER-EU framework.

Note

The TIBER-EU framework requires that the threat intelligence and the red team are performed by an independent third-party provider. This requirement ensures that an outside perspective is applied and may offer additional resources that internal teams may not have.

You can find additional information about the TIBER-EU framework at <https://www.ecb.europa.eu/paym/cyber-resilience/tiber-eu/html/index.en.html>.

CBEST

The Bank of England's Cyber Assessment Framework (CBEST) is another intelligence-led approach to red teaming. It is a comprehensive framework used to assess and enhance the cyber resilience of UK financial institutions. Red team activities are tailored to specific threats and vulnerabilities that are faced by the specific financial institution that is being assessed. It prioritizes the testing of critical systems and functions that are essential for the institution's operations. The assessment follows the principle of the "golden thread," which is a traceable link between the identified threats from the intelligence assessment and the resulting actions the firm takes to mitigate those threats. It ensures that there is alignment with the organization's overall risk management strategy, ensuring accountability and continuous improvement in their cybersecurity posture. It is common to utilize this framework if you are based in the UK or performing work there. An example would be a UK-based payment processor that wants to evaluate its defenses against data exfiltration. This type of framework would be required. In an exercise, the participants include four teams:

- **Threat Intelligence Service Provider:** This third-party team provides an external threat intelligence assessment, creates scenarios for the assessment, and completes the threat intelligence maturity assessment of the firm.
- **Penetration Test Service Provider:** This third-party team plans and executes based on the threat intelligence scenario, completes the detection & response capability assessment, drafts the penetration test findings report, and provides feedback.
- **Firm/Control Group:** The firm selects members for the control group. The control group coordinator (CGC) will coordinate all test activities

- for the firm to include project management of the assessment, quality assurance, and stakeholder coordination.
- **Regulator:** This team provides guidance and oversight throughout the assessment, reviews findings and remediation plans, and takes action on required regulatory activities.

Red Team Engagements

Red team engagements come in various forms, each tailored to specific objectives and organizational needs. Think of it like the right tool for the job. Choosing a hammer to turn a screw may not be the optimal outcome or effective use of resources. Similarly, different cybersecurity threats require different approaches to testing your defenses. The choice of engagement depends on a variety of factors, including the organization’s size and industry, its risk tolerance, and the desired outcomes it hopes to achieve. For example, a small startup will have different needs and risks when compared to a multinational bank. Similarly, an organization primarily concerned with data breaches will require different types of red team engagements than one focused on physical security vulnerabilities. Understanding the different types of red team engagements is crucial for selecting the most appropriate approach to assess and enhance an organization’s security posture. Let’s explore some of the most common types of red team engagements.

Full-Scope

A full-scope engagement is the classic and most extensive type of engagement. The red team utilizes everything within its arsenal, incorporating threat intelligence, physical and social engineering, as well as cyber, usually with an extended timeline to accomplish these activities. This approach allows for a holistic assessment of an organization and test of its resilience. During a full-scope engagement, a significant investment is required from a time and resource perspective. Depending on the size of the organization, this type of engagement may be too resource intensive to accomplish regularly.

Table 2-1 outlines the benefits and drawbacks of full-scope engagements.

Table 2-1 Full-Scope Engagement Benefits/Drawbacks

Benefits	Drawbacks
Holistic Assessment: The engagement assesses the organization’s overall resilience, including its ability to detect and respond to attacks across various domains.	Resource Intensive: These may require the development or purchase of new tools or services that are not regularly utilized by the team.

Benefits	Drawbacks
Multiple Attack Vectors: These can involve attacks on both digital systems as well as physical security, often combining multiple attacks for successful execution.	Long Duration: Engagements of these types can go on for months, while not requiring an entire team for the full duration of it. Aspects such as buildout and social engineering can last longer than a typical one.

Objective-Based

In objective-based engagements, the red team is given a clearly defined objective that aligns with a specific organizational goal or risk. The objective could be to access sensitive data (such as customer information or intellectual property), to manipulate and/or exfiltrate the data. These objective-based engagements typically have a defined start and end. Due to the scope being focused on achieving a specific goal, these engagements tend to be shorter than full-scope engagements. However, they can still vary from a week to several months depending on the complexity.

Table 2-2 outlines the benefits and drawbacks of objective-based engagements.

Table 2-2 Objective-Based Engagement Benefits/Drawbacks

Benefits	Drawbacks
Clear and Measurable Outcome: These engagements have a specific outcome that can be easily determined if it is successful and can be measured.	Narrow Scope: By focusing on a single objective, these engagements may overlook other critical vulnerabilities or risks within the organization.
Efficient and Resource Optimization: With the narrower focus and clear mission of this engagement, the resources required are easier to define.	Overreliance on Predefined Outcomes: Success or failure is measured against achieving a specific goal, potentially leading to a “binary” evaluation of the engagement’s results.
Alignment with Business Risks: The engagement’s objective is chosen to reflect the organization’s specific risks, which makes it more impactful. Whether testing for data breaches, physical security, or ransomware resilience, each exercise aligns with the highest-priority risks the organization faces.	Dependency on Proper Objective Selection: The success of the engagement depends heavily on defining an appropriate and relevant objective. If the chosen objective does not align with the organization’s real-world risks, the exercise may provide little value. Poorly selected objectives can lead to a misallocation

Benefits	Drawbacks
	of resources and a false sense of security.
Improved Incident Response and Detection/Alerting: Objective-based exercises test the organization's ability to detect and respond to a specific type of threat. By focusing on a particular attack scenario, the blue team can improve its incident response capabilities in a focused way.	
Adaptability: With a defined objective, red team engagements can quickly adapt to emerging threats, such as new ransomware techniques or supply chain attacks. This flexibility enables the organization to test its defenses against the latest, most relevant adversarial tactics.	

Adversary Emulation (Scenario-Based)

In adversary emulation engagements, the focus is on simulating a realistic, end-to-end attack scenario that mimics the tactics, techniques, and procedures of a specific threat actor or type of adversary. Unlike objective-based engagements that have a narrowly defined goal, adversary emulation-based engagements take a broader approach, replicating the lifecycle of an attack from initial reconnaissance to achieving multiple potential outcomes. The scenarios are designed to represent real-world attack conditions, emphasizing the organization's ability to defend against, detect, and respond to evolving threats. These engagements are often rooted in specific use cases, such as advanced persistent threat (APT) simulations, supply chain attacks, or ransomware incidents, and can range from a few weeks to several months, depending on the complexity and scope. The use of threat intelligence is critical in determining the relevant threat actor and TTPs to emulate.

Table 2-3 outlines the benefits and drawbacks of adversary emulation (scenario-based) engagements.

Table 2-3 Adversary Emulation (Scenario-Based) Engagement Benefits/Drawbacks

Benefits	Drawbacks
Comprehensive Threat Emulation: This engagement provides a more holistic view of the organization's security posture by	Broad Scope: The expansive nature of these engagements can dilute focus, potentially overlooking critical vulnerabili-

Benefits	Drawbacks
emulating a full attack lifecycle based on specific threats.	ties that might have been identified in more targeted testing.
Real-World Relevance: These engagements reflect current and emerging threats, incorporating TTPs associated with active threat actors or industry-specific risks.	Difficulty Measuring Outcomes: Unlike objective-based engagements, where success is tied to a single goal, scenario-based engagements have multiple outcomes, making it harder to measure success.

Purple Team

These types of exercises involve a collaborative approach where both the red team and blue team work together to simulate and respond to attacks in real time. The goal is mutual learning and continuous improvement rather than just adversarial exercises. The red team demonstrates and executes emulated attacks, while the blue team actively detects, responds to, and mitigates these attacks. This engagement fosters open communication and knowledge sharing, allowing both teams to understand each other’s perspectives and enhance the organization’s overall security posture.

Table 2-4 outlines the benefits and drawbacks of purple team engagements.

Table 2-4 Purple Team Engagement Benefits/Drawbacks

Benefits	Drawbacks
Real-Time Collaboration: Red and blue teams work together, with the red team demonstrating the attack while the blue team practices detection and response.	Resource Intensive: An effective purple team requires experienced personnel from both teams, coordination, and time to conduct meaningful exercises.
Immediate Feedback: This approach allows for immediate feedback, improving security controls during the exercise rather than in post-exercise analysis.	Knowledge Silos: Without proper documentation, reporting, and cross-training of personnel, the knowledge gained during an exercise may not be disseminated beyond the participants.

Benefits	Drawbacks
Continuous Improvement: Purple teaming fosters a culture of continuous improvement, enabling both teams to share insights and improve overall security posture iteratively.	Potential for Reduced Realism: When both teams are aware of the attacks being simulated, the response may not be fully executed and tested. This can be due to assumptions and usage of a testing environment that doesn't replicate production.

Tabletop

Tabletop exercises (TTXs) are structured, discussion-based security simulations designed to evaluate an organization's readiness for handling various scenarios. These scenarios can take on a wide variety of types, from cyber attacks to natural disasters or other crises. Unlike live simulations, TTXs are conducted in a controlled environment where participants discuss actions and responses to hypothetical incidents without actively deploying tools or systems. These exercises are typically facilitated by a moderator who guides the scenario, posing challenges and prompting participants to consider their response strategies. TTXs focus on testing policies, processes, communication, and decision-making rather than technical capabilities. They are valuable for identifying gaps in incident response plans, fostering cross-functional collaboration, and enhancing overall preparedness. Depending on the organization, a TTX is something that the red team may participate in or facilitate; however, it is something that is often provided by a third party.

Table 2-5 outlines the benefits and drawbacks of TTX engagements.

Table 2-5 TTX Benefits/Drawbacks

Benefits	Drawbacks
Identifying Gaps in Plans and Processes: TTXs expose weaknesses in existing policies, playbooks, and communication protocols, allowing organizations to address them proactively.	Overlooking Technical Details: TTXs primarily test processes and policies, which can leave technical gaps unidentified.
Improved Communication and Collaboration: Through involving multiple teams, TTXs strengthen interdepartmental communication and clarify roles and responsibilities during a crisis.	Time Constraints: Limited availability of participants from across the organization may limit participation of the depth of the exercise.
Cost-Effective: Because TTXs do not involve active systems or tools, they are less resource-intensive than live simulations while still offering valuable insights. However, keep in mind the cost	

Benefits	Drawbacks
of the personnel involved in the exercise.	

Red Team Challenges

There are a number of challenges that red teamers will face during their career. While they don't happen often, it is important to understand and prepare for them when they do arise. The following sections explore some of the common challenges faced by red teams, from navigating complex organizational structures to dealing with the frustration of "no-findings" engagements. By acknowledging these hurdles and developing strategies to overcome them, red teams can consistently improve their effectiveness and contribute to a stronger security posture for the organization. Let's explore some of those challenges that red teams may face.

Leveling Up Other Teams

An often overlooked aspect of red teaming is delivering value to other teams. It's not just about identifying vulnerabilities and doing cool things; the true value is that the security knowledge across the organization has risen. A mature red team can contribute to other teams in numerous ways.

A red team can significantly boost the blue team's capabilities by providing them with realistic attack scenarios and insights into adversary mindsets. This practical experience allows the blue team to improve their detection and response mechanisms, fine-tune security tools, and strengthen their overall incident response capabilities. Essentially, the red team acts as a sparring partner, helping the blue team to train and prepare for real-world threats.

The red team's findings provide invaluable feedback for the threat intelligence team. Through sharing information on exploited vulnerabilities, attack vectors, and observed tactics, techniques, and procedures, the red team helps the threat intelligence team refine their threat models, improve threat-hunting efforts, and provide more accurate and timely warnings within the organization. This collaborative loop ensures that the organization's understanding of the threat landscape is continuously updated and aligned with the latest threats.

Red team engagements can be leveraged to create compelling security awareness training for employees. By demonstrating real-world attack scenarios and their potential impact, the red team can educate users about common threats like phishing, social engineering, and malware. This practical approach helps employees understand the importance of security protocols and encourages them to adopt secure behaviors, ultimately reducing the organization's vulnerability. This approach can take on many forms from something as simple as a "brown bag lunch demo" to a presentation during an internal conference. A more hands-on and focused approach will be discussed in the next paragraph.

By collaborating with engineering teams, red teams can provide insights into secure coding practices and design considerations. This effort can in-

involve vulnerability assessments of new applications and threat modeling during the development lifecycle. This is not the red team performing this function but providing expertise if it is necessary. Nor does this replace formal training in these specialties, but it provides just-in-time and proactive guidance should the need arise. An example would be the identification of inconsistent user input sanitization in various portions of products. The red team can collaborate with engineers to identify the best practices that are relevant to the organization and then work to disseminate that information in the appropriate manner with the relevant engineers.

Lack of Findings

Not every engagement is going to be perfect or full of the latest vulnerabilities. There will be times when it may seem like a failure. Not achieving your objective is just as informative. You will have performed a validation of existing security controls. It proves that they have been configured and implemented correctly, but maybe this is a surprise for you. In the last couple of engagements, you just steamrolled through them; however, this time, they closed out that Jira ticket for tweaking the control. There may have been an implementation of a new security control that the team was not tracking. This process provides an opportunity for the team to learn and experiment with security control. Finally, it provides guidance for future engagements. When the team is planning the next set of engagements, there is an opportunity to refine the scope, re-evaluate risk, and define new objectives. It is an opportunity to learn and improve, not a sign of defeat.

This chapter introduces red teaming, highlighting its significance in cybersecurity and contrasting it with penetration testing. While penetration tests focus on uncovering vulnerabilities in a controlled scope, red teaming adopts a broader, adversarial approach to assess an organization's overall resilience, including people, processes, and technology.

We then discussed the importance of red teaming in identifying critical vulnerabilities, challenging security assumptions, and simulating industry-specific threats. The chapter underscores how adversarial perspectives foster realistic security measures and adaptive defenses.

We also explored key frameworks and methodologies, including

- **MITRE ATT&CK Framework:** A detailed catalog of adversarial tactics and techniques
- **Unified Kill Chain:** A comprehensive view of attack lifecycles tied to adversary goals
- **TIBER-EU and CBEST:** Frameworks tailored for financial institutions, focusing on threat intelligence and consistent testing standards

Various types of red team engagements are explained:

- **Full-Scope:** Holistic assessments involving multiple attack vectors
- **Objective-Based:** Assessments focused on achieving specific, measurable goals
- **Adversary Emulation:** Simulations of realistic, end-to-end attack scenarios based on threat intelligence

- **Purple Teaming:** Collaborative exercises between red and blue teams for real-time learning and improvement
- **Tabletop Exercises:** Discussion-based simulations to evaluate incident readiness

Finally, we emphasized how red teaming enhances detection, response, and recovery capabilities, fostering a resilient security posture aligned with modern threats.

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Test Your Skills

Multiple-Choice Questions

1. Which of the following best describes the mindset encouraged by red teaming?
 1. Defensive
 2. Risk-taking
 3. Adversarial
 4. Observant
2. Which framework is a comprehensive catalog of TTPs of adversaries?
 1. TIBER-EU
 2. CBEST
 3. Unified Kill Chain
 4. MITRE ATT&CK
3. Why is it important for red teams to emulate adversaries effectively?
 1. To demonstrate technical skills
 2. To validate existing assumptions
 3. To challenge assumptions of security controls
 4. To complete exercises quickly
4. How does red teaming contribute to an organization’s resilience?
 1. Through enforcing strict adherence to policies
 2. By revealing potential risks and weaknesses
 3. Through reducing costs in strategic planning
 4. By prioritizing immediate implementation of changes
5. What is a drawback of purple teaming?
 1. Resource intensive
 2. Difficulty measuring outcome
 3. Long duration
 4. Adaptability
6. You are executing an engagement with a UK-based financial institution. What framework might you be required to use?
 1. TIBER-EU
 2. CBEST
 3. Unified Kill Chain
 4. MITRE ATT&CK